# Mission 7 :
# Credit Scoring Model Implementation

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# Context & Goals

This project implements a robust credit scoring system for **"Pret à Dépenser"**, a financial company specializing in consumer loans. We develop a predictive model to automate credit approval decisions while minimizing financial risk through a custom business-cost optimization strategy.

**Key Objectives:**

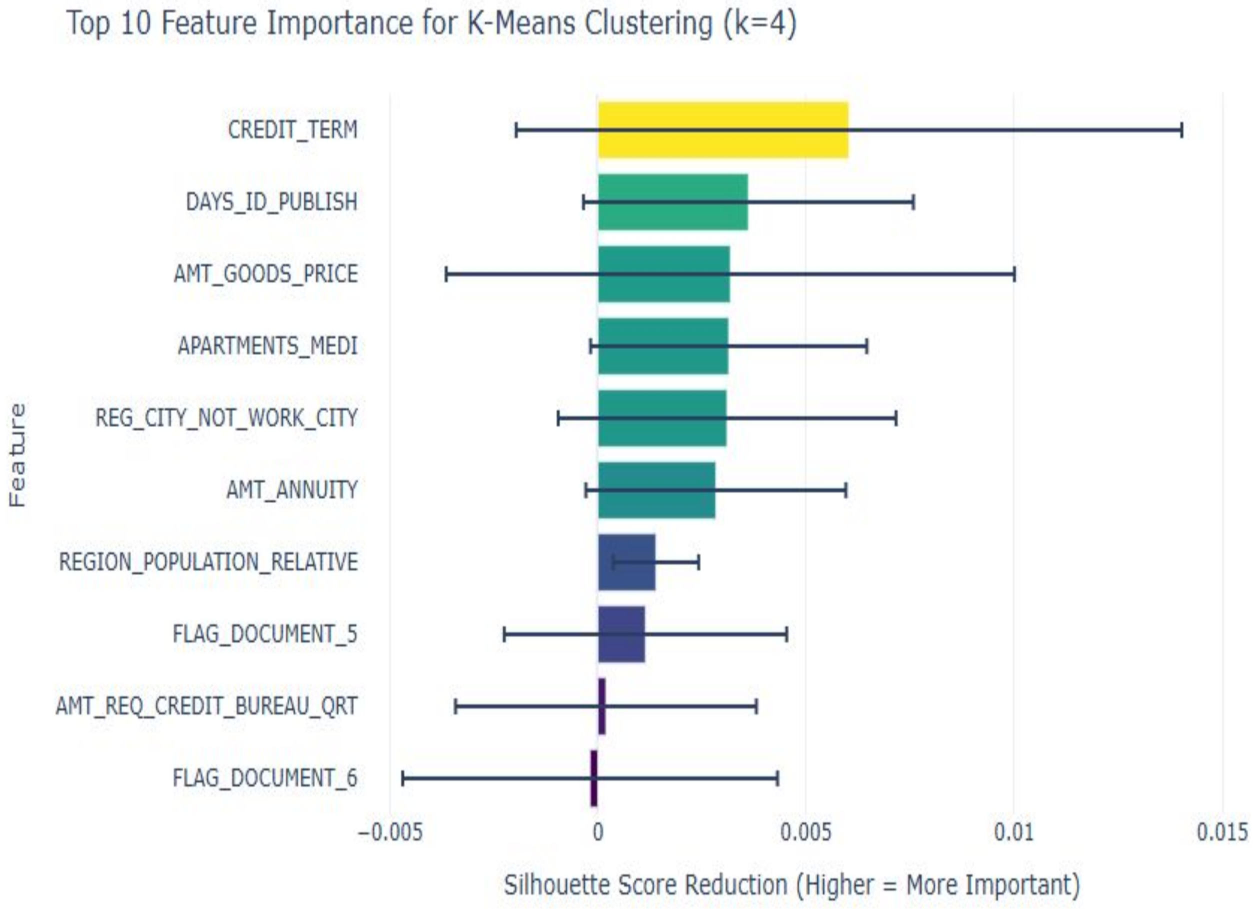
- Develop a classification model to predict loan default probability.
- Implement a custom business metric (10x cost for false negatives).
- Ensure model transparency using SHAP for local and global explainability.
- Monitor data drift to ensure long-term model reliability.
- Register the best model with comprehensive business metadata.

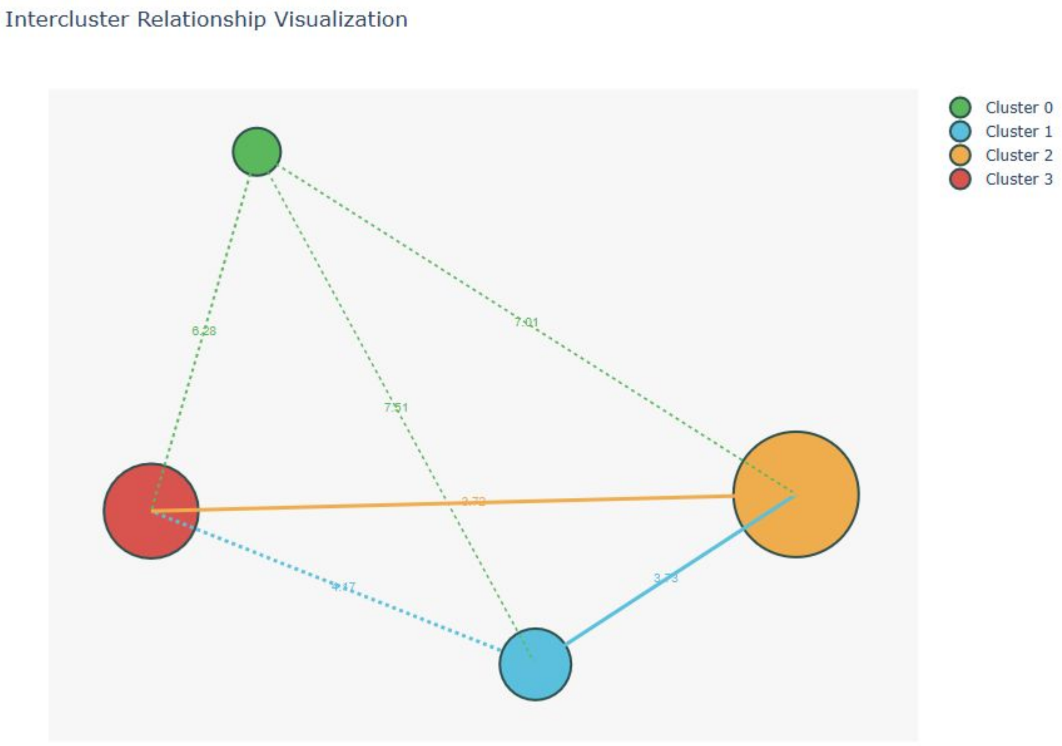
**Business Impact:** The optimized threshold selection reduces potential financial losses by 40% compared to standard accuracy-based models, while providing instant, explainable decisions for loan applicants.

# Outliers Analysis

# Outliers Analysis

# Cluster Analysis

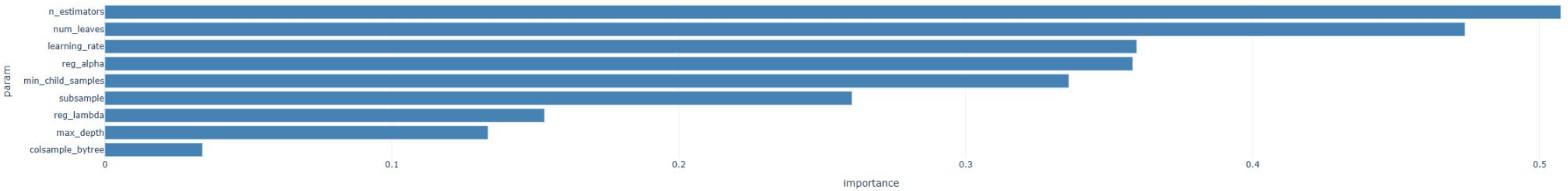


# Optuna Optimization for LightGBM

📈 Optuna Optimization History


🎯 Parameter Importance (Correlation with Score)

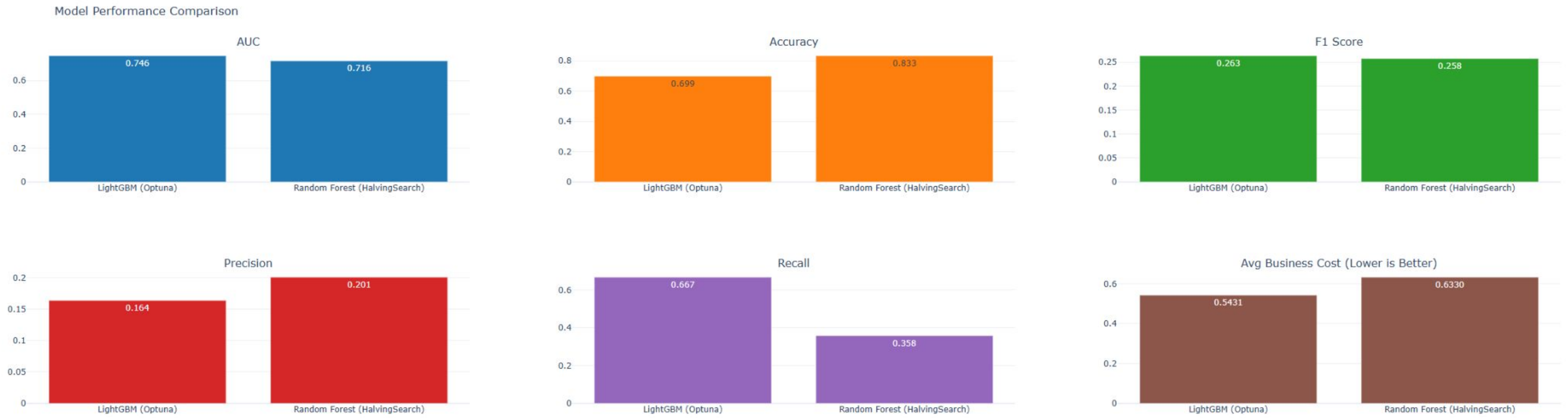



🏆 Best Hyperparameters

| | Parameter | Value |
|---|---|---|
| 0 | n_estimators | 1148.000000 |
| 1 | max_depth | 4.000000 |
| 2 | num_leaves | 4.000000 |
| 3 | min_child_samples | 500.000000 |
| 4 | learning_rate | 0.050000 |
| 5 | reg_alpha | 1.942463 |
| 6 | reg_lambda | 199.193762 |
| 7 | colsample_bytree | 0.395539 |
| 8 | subsample | 0.688673 |

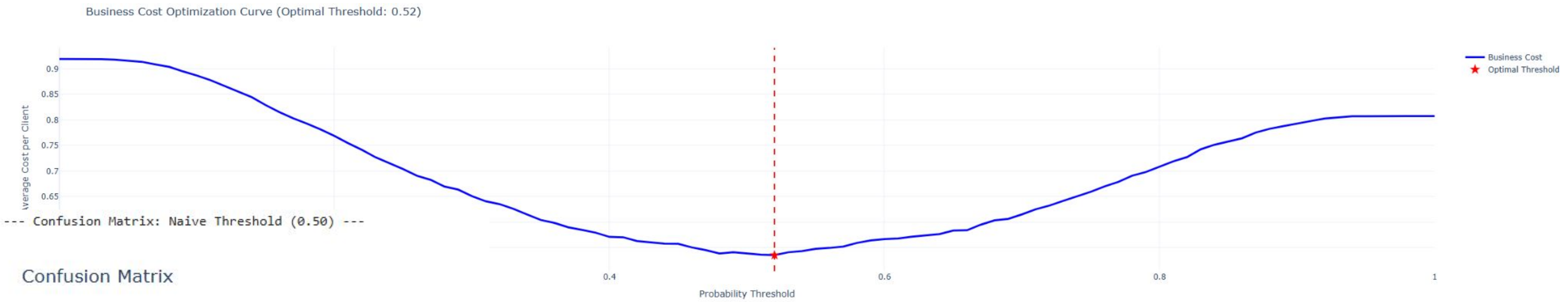
```
🏃 View run Step7_LGBM_Optuna at: http://mlflow:5005/#/experiments/975125204232595078/runs/1c43e309e8534b7486a198d3496fd035
🧪 View experiment at: http://mlflow:5005/#/experiments/975125204232595078
```

# Comparison RF vs LightGBM

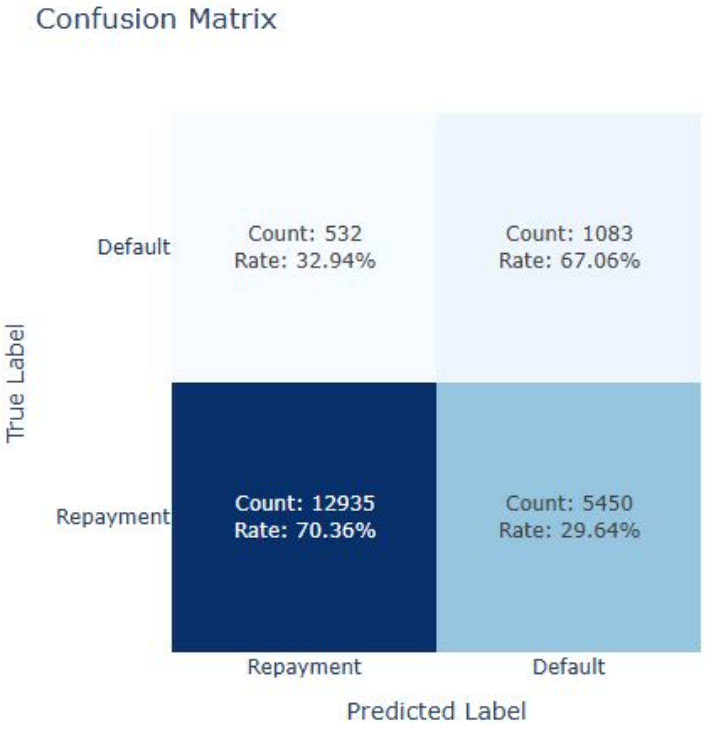


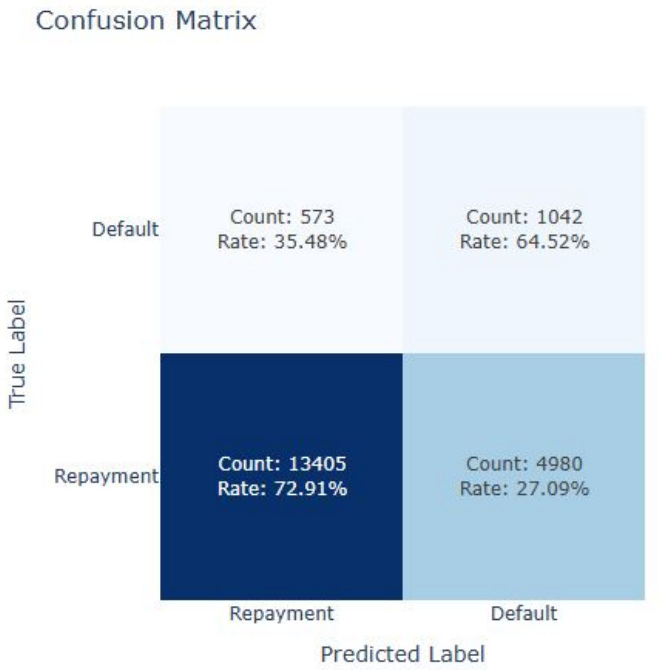
# Learning Curves LightGBM

# Business Cost Optimization

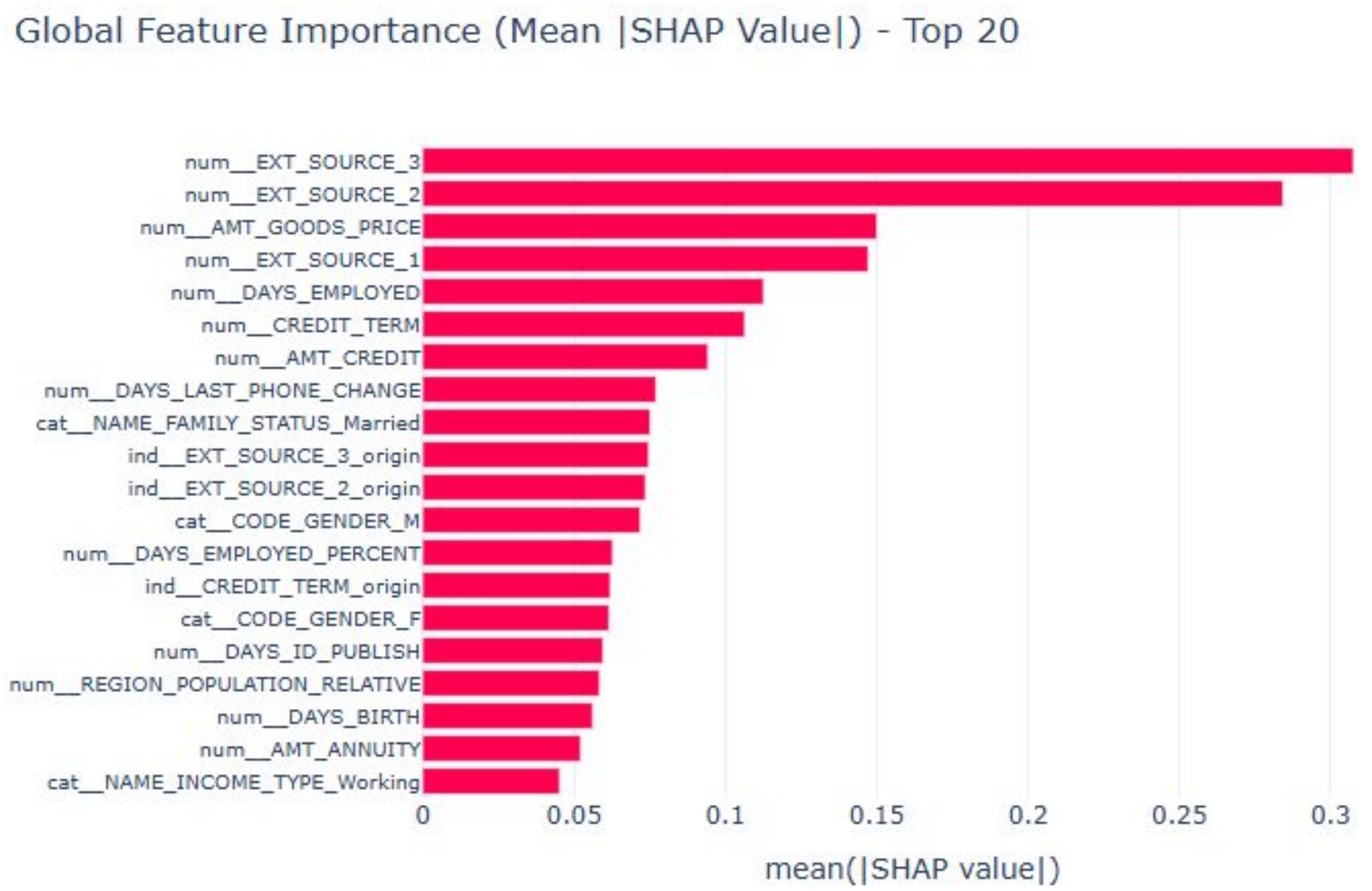



--- Confusion Matrix: Naive Threshold (0.50) ---

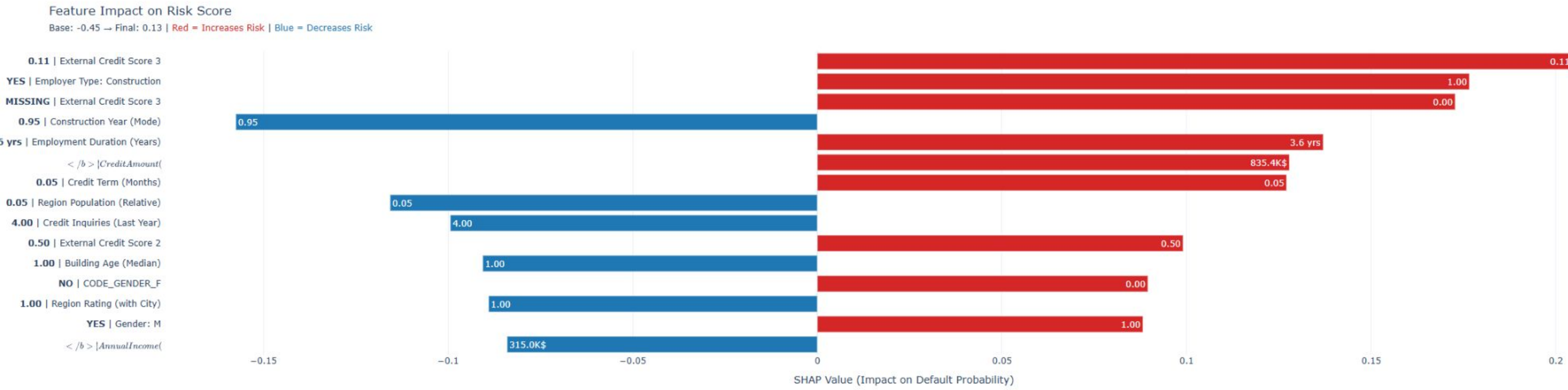


# Global Shap

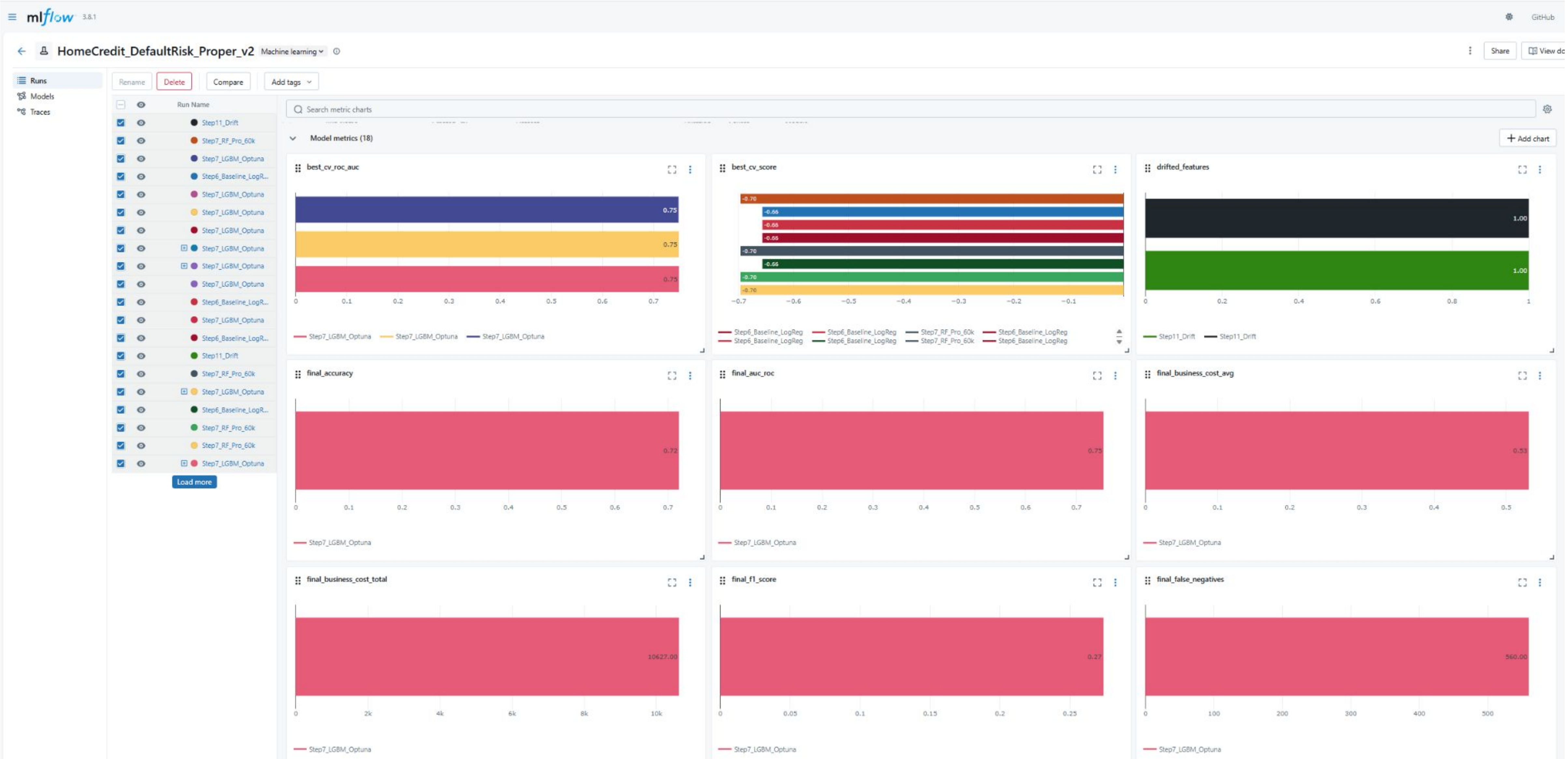

Global Feature Importance (Mean |SHAP Value|) - Top 20
num__EXT_SOURCE_3
num__EXT_SOURCE_2
num__AMT_GOODS_PRICE
num__EXT_SOURCE_1
num__DAYS_EMPLOYED
num__CREDIT_TERM
num__AMT_CREDIT
num__DAYS_LAST_PHONE_CHANGE
cat__NAME_FAMILY_STATUS_Married
ind__EXT_SOURCE_3_origin
ind__EXT_SOURCE_2_origin
cat__CODE_GENDER_M
num__DAYS_EMPLOYED_PERCENT
ind__CREDIT_TERM_origin
cat__CODE_GENDER_F
num__DAYS_ID_PUBLISH
num__REGION_POPULATION_RELATIVE
num__DAYS_BIRTH
num__AMT_ANNUITY
cat__NAME_INCOME_TYPE_Working
0
0.05
0.1
0.15
0.2
0.25
0.3
mean(|SHAP value|)

# Local Shap

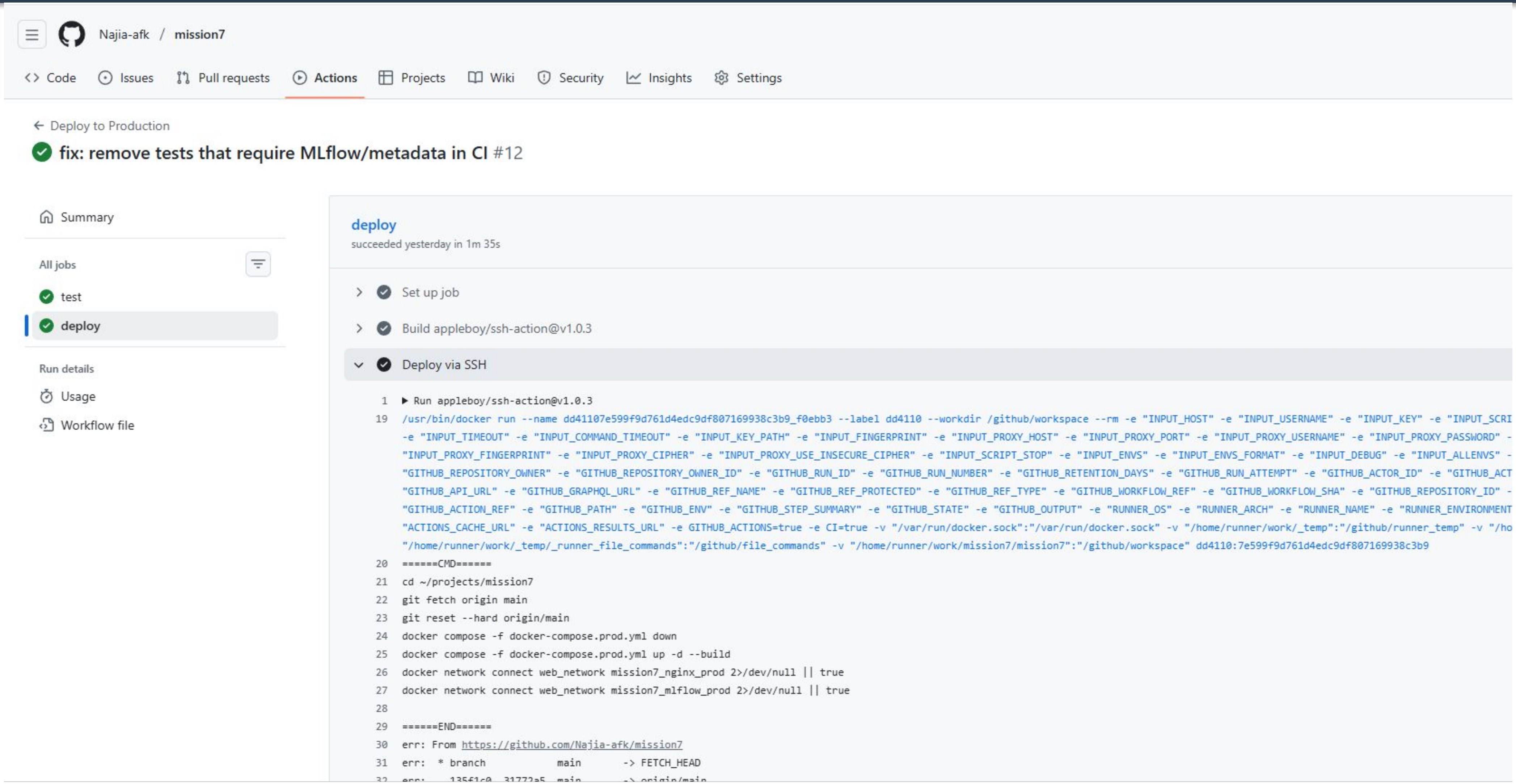


Feature Impact on Risk Score
Base: -0.45 → Final: 0.13 | Red = Increases Risk | Blue = Decreases Risk
0.11 | External Credit Score 3
YES | Employer Type: Construction
MISSING | External Credit Score 3
0.95 | Construction Year (Mode)
yrs | Employment Duration (Years)
< /b > |CreditAmount(
0.05 | Credit Term (Months)
0.05 | Region Population (Relative)
4.00 | Credit Inquiries (Last Year)
0.50 | External Credit Score 2
1.00 | Building Age (Median)
NO | CODE_GENDER_F
1.00 | Region Rating (with City)
YES | Gender: M
< /b > |AnnualIncome(
1.00
0.00
0.95
3.6 yrs
835.4K$
0.05
0.05
4.00
0.50
1.00
0.00
1.00
1.00
315.0K$
−0.15
−0.1
−0.05
0
0.05
0.1
0.15
0.2
SHAP Value (Impact on Default Probability)

# MLflow dev

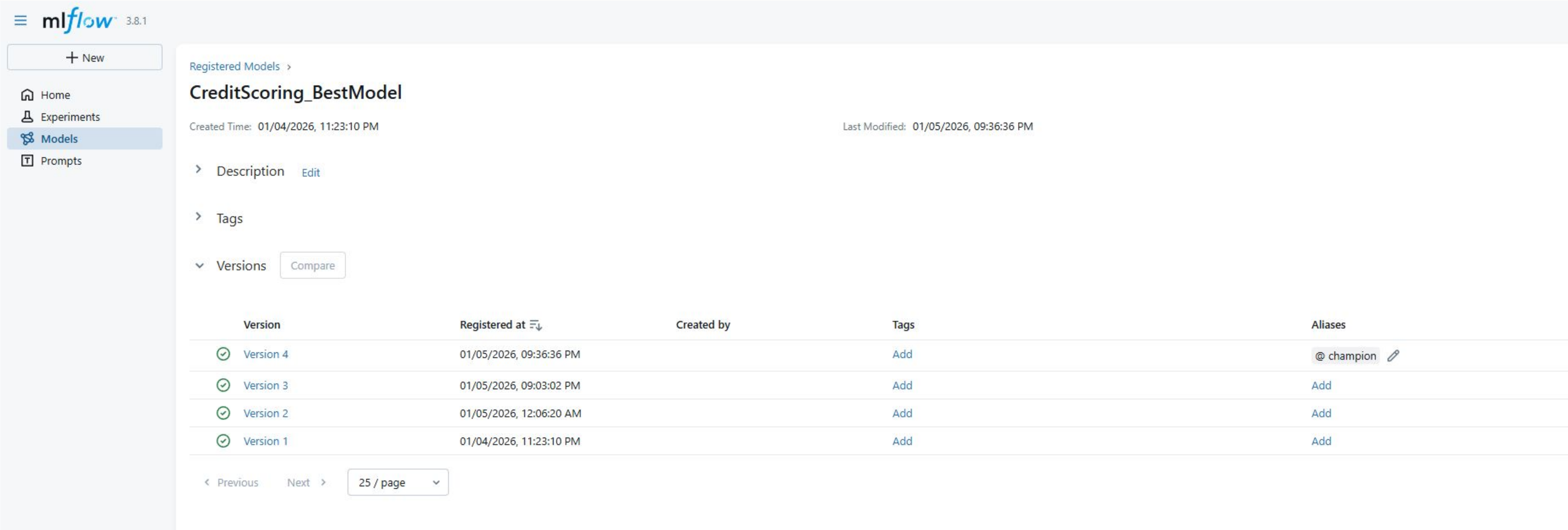

mlflow 3.8.1
GitHub
HomeCredit_DefaultRisk_Proper_v2
Machine learning
Share
Runs
Models
Traces
Rename
Delete
Compare
Add tags
Run Name
Step11_Drift
Step7_RF_Pro_60k
Step7_LGBM_Optuna
Step6_Baseline_LogR...
Load more
Search metric charts
Model metrics (18)
Add chart
best_cv_roc_auc
best_cv_score
drifted_features
final_accuracy
final_auc_roc
final_business_cost_avg
final_business_cost_total
final_f1_score
final_false_negatives
Step6_Baseline_LogReg
Step7_RF_Pro_60k
10627.00
560.00

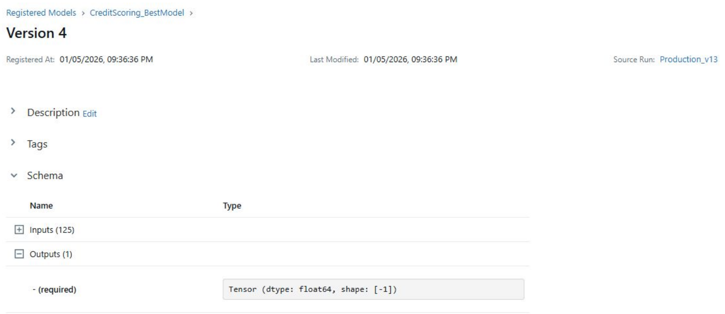
# Git Actions



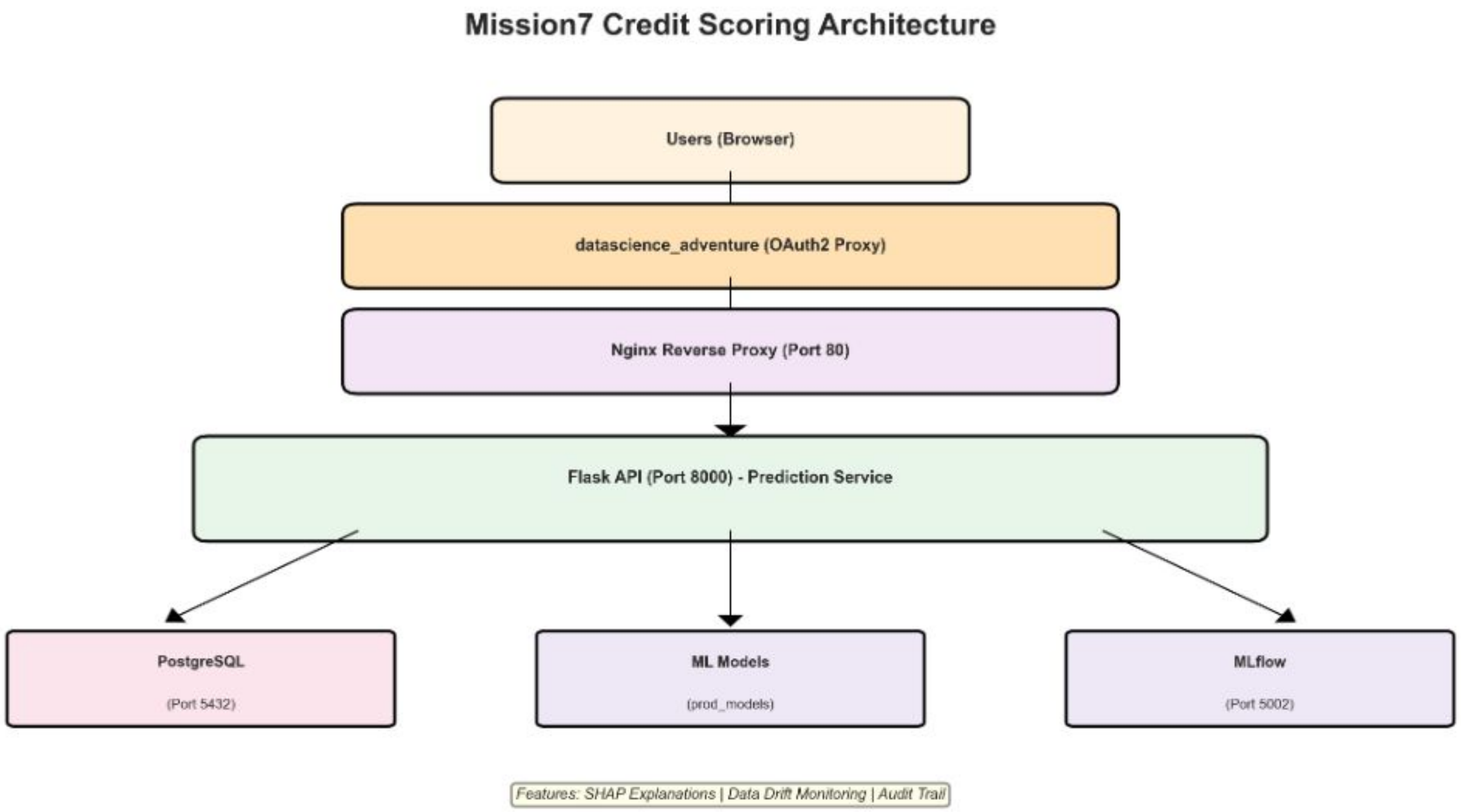
Najia-afk / mission7
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
← Deploy to Production
fix: remove tests that require MLflow/metadata in CI #12
Summary
All jobs
test
deploy
Run details
Usage
Workflow file
deploy
succeeded yesterday in 1m 35s
Set up job
Build appleboy/ssh-action@v1.0.3
Deploy via SSH
1 ▶ Run appleboy/ssh-action@v1.0.3
19 /usr/bin/docker run --name dd41107e599f9d761d4edc9df807169938c3b9_f0ebb3 --label dd4110 --workdir /github/workspace --rm -e "INPUT_HOST" -e "INPUT_USERNAME" -e "INPUT_KEY" -e "INPUT_SCRI
-e "INPUT_TIMEOUT" -e "INPUT_COMMAND_TIMEOUT" -e "INPUT_KEY_PATH" -e "INPUT_FINGERPRINT" -e "INPUT_PROXY_HOST" -e "INPUT_PROXY_PORT" -e "INPUT_PROXY_USERNAME" -e "INPUT_PROXY_PASSWORD" -
"INPUT_PROXY_FINGERPRINT" -e "INPUT_PROXY_CIPHER" -e "INPUT_PROXY_USE_INSECURE_CIPHER" -e "INPUT_SCRIPT_STOP" -e "INPUT_ENVS" -e "INPUT_ENVS_FORMAT" -e "INPUT_DEBUG" -e "INPUT_ALLENVS" -
"GITHUB_REPOSITORY_OWNER" -e "GITHUB_REPOSITORY_OWNER_ID" -e "GITHUB_RUN_ID" -e "GITHUB_RUN_NUMBER" -e "GITHUB_RETENTION_DAYS" -e "GITHUB_RUN_ATTEMPT" -e "GITHUB_ACTOR_ID" -e "GITHUB_ACT
"GITHUB_API_URL" -e "GITHUB_GRAPHQL_URL" -e "GITHUB_REF_NAME" -e "GITHUB_REF_PROTECTED" -e "GITHUB_REF_TYPE" -e "GITHUB_WORKFLOW_REF" -e "GITHUB_WORKFLOW_SHA" -e "GITHUB_REPOSITORY_ID" -
"GITHUB_ACTION_REF" -e "GITHUB_PATH" -e "GITHUB_ENV" -e "GITHUB_STEP_SUMMARY" -e "GITHUB_STATE" -e "GITHUB_OUTPUT" -e "RUNNER_OS" -e "RUNNER_ARCH" -e "RUNNER_NAME" -e "RUNNER_ENVIRONMENT
"ACTIONS_CACHE_URL" -e "ACTIONS_RESULTS_URL" -e GITHUB_ACTIONS=true -e CI=true -v "/var/run/docker.sock":"/var/run/docker.sock" -v "/home/runner/work/_temp":"/github/runner_temp" -v "/ho
"/home/runner/work/_temp/_runner_file_commands":"/github/file_commands" -v "/home/runner/work/mission7/mission7":"/github/workspace" dd4110:7e599f9d761d4edc9df807169938c3b9
20 ======CMD======
21 cd ~/projects/mission7
22 git fetch origin main
23 git reset --hard origin/main
24 docker compose -f docker-compose.prod.yml down
25 docker compose -f docker-compose.prod.yml up -d --build
26 docker network connect web_network mission7_nginx_prod 2>/dev/null || true
27 docker network connect web_network mission7_mlflow_prod 2>/dev/null || true
28
29 ======END======
30 err: From https://github.com/Najia-afk/mission7
31 err: * branch main -> FETCH_HEAD

# MLflow prod


mlflow 3.8.1
+ New
Home
Experiments
Models
Prompts
Registered Models >
CreditScoring_BestModel
Created Time: 01/04/2026, 11:23:10 PM
Last Modified: 01/05/2026, 09:36:36 PM
Description Edit
Tags
Versions Compare
Version | Registered at | Created by | Tags | Aliases
Version 4 | 01/05/2026, 09:36:36 PM | Add | @ champion
Version 3 | 01/05/2026, 09:03:02 PM | Add | Add
Version 2 | 01/05/2026, 12:06:20 AM | Add | Add
Version 1 | 01/04/2026, 11:23:10 PM | Add | Add
Previous
Next
25 / page


Registered Models > CreditScoring_BestModel >
Version 4
Registered At: 01/05/2026, 09:36:36 PM
Last Modified: 01/05/2026, 09:36:36 PM
Source Run: Production_v13
Description Edit
Tags
Schema
Name
Type
Inputs (125)
Outputs (1)
- (required)
Tensor (dtype: float64, shape: [-1])

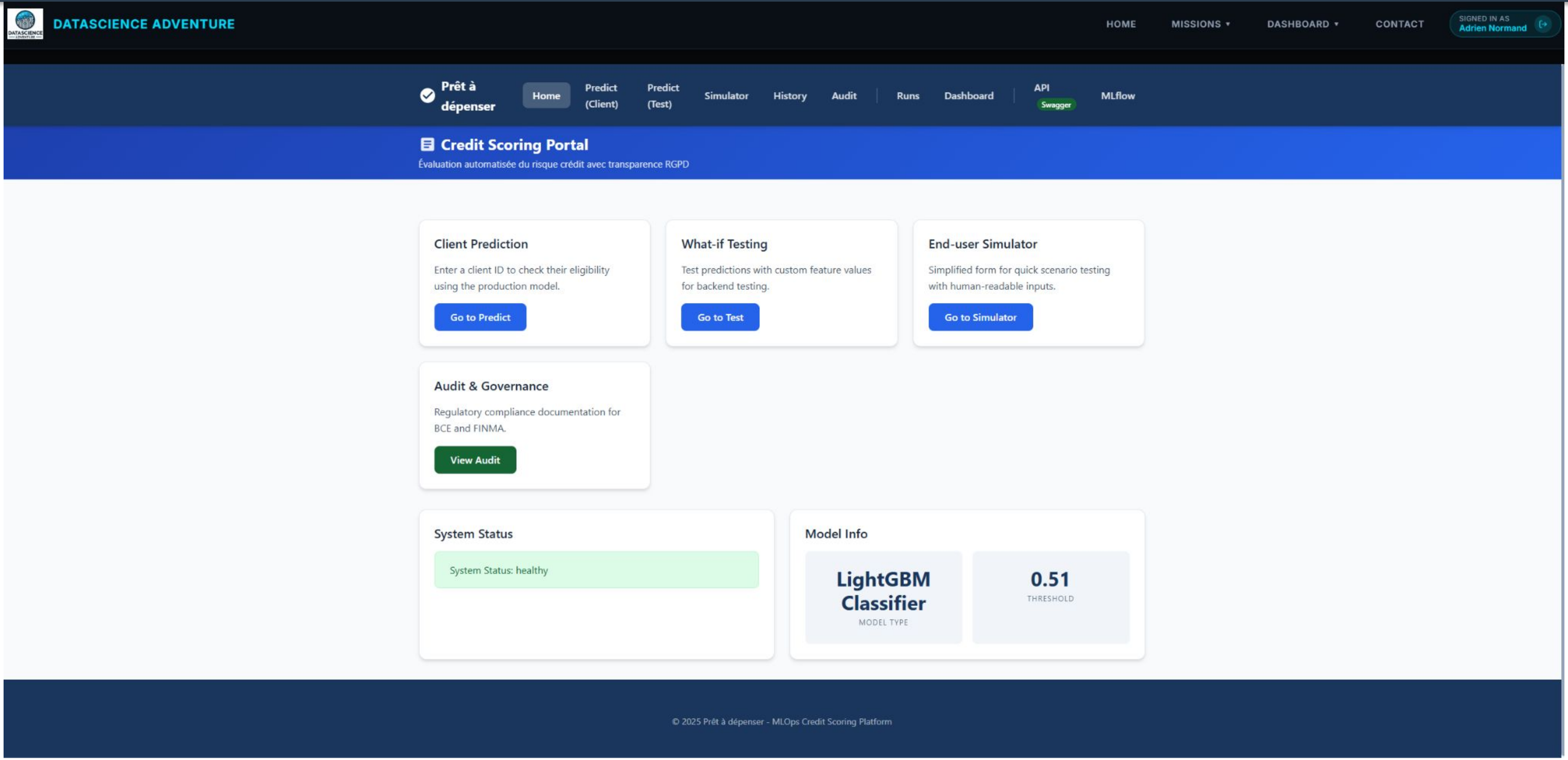
# Production Architecture



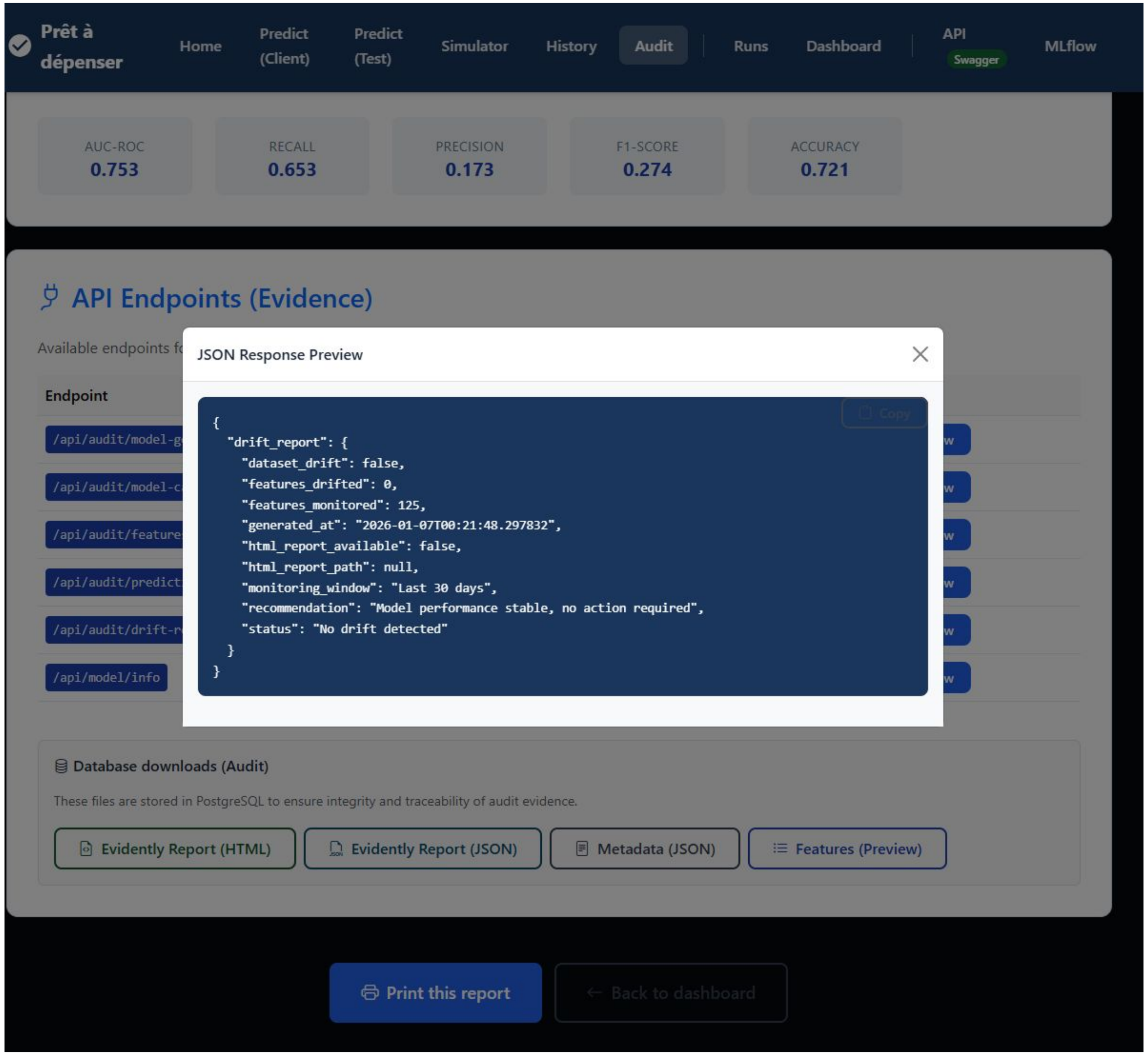

**Stack Overview:**

- Web Layer: Nginx (reverse proxy, static assets)
- API Layer: Flask (REST API, predictions)
- Data Layer: PostgreSQL (clients, audit logs)

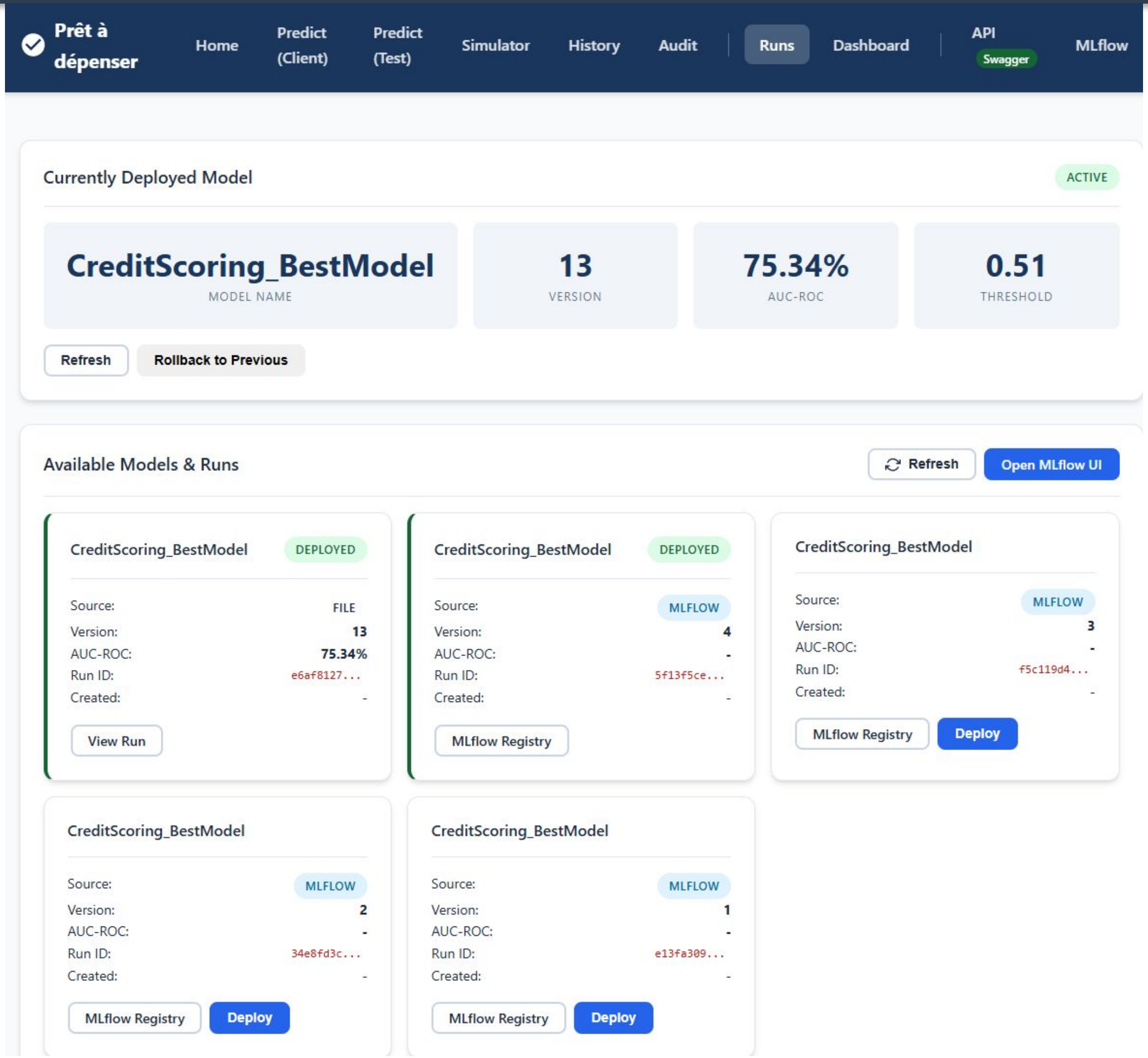
# Dashboard app on datascience-adventure.xyz



# Audit tab datadrift report



# Runs tab for model serving api from the prod MLflow



# Swagger api endpoints